Learned Acoustic Signalling for Occlusion-Aware Robot Navigation in Dense Crowds

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***Abstract*—Can a crowd-navigation robot use sound both to clear its path and to locate people it cannot see? We give a deep reinforcement learning policy a learned acoustic action: it can beep, nearby pedestrians may step aside and reply, and the robot hears the replies as noisy range and bearing estimates, including from pedestrians hidden behind obstacles. The policy builds on HEIGHT, a heterogeneous graph-transformer navigation policy, extended with trajectory prediction and variational occlusion inference, and is trained with proximal policy optimisation in a PyBullet simulation with a differential-drive TurtleBot2 among ORCA pedestrians. In a matched-budget ablation evaluated on $3{,}500$ episodes per arm, the beep raises success by about one point at the nominal crowd density (not significant on one test set) and by 2.5 points with 10–14 pedestrians ($p = 0.008$). Hearing the replies localises 2.6 hidden pedestrians per episode but adds nothing measurable to success over beeping alone: the benefit comes from clearing the path, not from sensing. It also depends on a simulator assumption we state explicitly, that pedestrians who cannot perceive the robot still obey its beep. We report the design, the reward, the failures we found along the way, and the size of each model.**

I. Introduction

Picture a delivery robot in a hospital corridor at shift change. Most people see it and step aside; one nurse reading a chart does not, and the robot is struck from the side. The scenario contains the central difficulty of social navigation: the robot's safety depends on behaviour it does not control, and on people it cannot always see.

This project gives a learned navigation policy a way to *act on the crowd*. Starting from a platform built on HEIGHT [1], a current reference architecture for crowd navigation, extended with a module that anticipates where each pedestrian will be over the next half second [2] and a module that infers pedestrians hidden behind obstacles [3], we gave the policy a beep, learned jointly with its motion. Nearby pedestrians may answer by stepping aside and by replying, so that the robot can also localise people it cannot see by ear.

Objectives. (i) Design an acoustic signalling and listening extension and train it with deep reinforcement learning; (ii) separate its two possible mechanisms, clearing the path and sensing hidden pedestrians, with matched-budget ablations; (iii) evaluate every variant at a statistical power high enough to detect a two-point change in success, including in denser crowds and among more obstacles; (iv) report model size and training compute, which matter for running such a policy on an on-board computer. The base platform's own evaluation against HEIGHT is the subject of a companion study [8]; here it serves as the matched baseline.

The results were mostly negative, and we report them in full. Section III states the learning problem, Sections IV and V describe the system and the acoustic extension, Section VI reports the evaluation, and Sections VII and VIII discuss what the results do and do not show.

II. Background

Crowd navigation with deep RL. A robot must reach a goal through a space containing moving pedestrians and static obstacles. Learned approaches encode the scene as a graph—robot, humans and obstacles as nodes, interactions as edges—and train a policy by reinforcement learning in simulation [1], [2], [4]. HEIGHT [1] distinguishes human and obstacle nodes and applies separate attention networks to robot-human and human-human edges.

Pedestrian simulation. Pedestrians follow ORCA [5], a reciprocal velocity-obstacle method: each agent selects a velocity that avoids every agent it observes. HEIGHT's configuration sets `robot.visible_prob` $= 0.2$: each moving pedestrian has a 20% chance of including the robot in its ORCA observation, and standing pedestrians never do, so averaged over the crowd-size distribution 17.6% of pedestrians can perceive the robot. We keep this benchmark setting throughout.

Occlusion inference. Mun *et al.* [3] train two variational encoders: a teacher on ground-truth occupancy and a student on partial lidar occupancy, with a distillation loss pulling the student's latent toward the teacher's. At inference only the student runs.

Acoustic signalling. Nishimura and Yonetani [6] let a robot learn when to emit an audible cue that makes nearby pedestrians yield, trading safety against efficiency. Our beep model follows theirs; the listening channel, in which pedestrians' replies are used as a sensor, is our addition.

III. Learning Problem

We treat navigation as a partially observed Markov decision process with time step $\Delta t = 0.1\,\mathrm{s}$ and a $50\,\mathrm{s}$ time limit in a

12×12 m arena with 5–9 pedestrians and 8–12 rectangular obstacles per episode.

Observation. The robot state $[p_x, p_y, g_x, g_y, \theta]$; for each detected pedestrian its position and velocity, extended by predicted future positions (Sec. IV); a 180-ray lidar scan of static obstacles; and an occupancy grid of visible pedestrians for occlusion inference. Pedestrians hidden behind obstacles or other pedestrians are not observed.

Actions. Nine discrete changes of commanded linear and angular velocity, $(\Delta v, \Delta\omega) \in \{-0.05, 0, 0.05\}\,\mathrm{m/s} \times \{-0.1, 0, 0.1\}\,\mathrm{rad/s}$, with $v \leq 0.5$ m/s. With the acoustic extension the action space doubles to 18: each motion with or without a beep.

Reward.

$$r_t = r_{\text{goal}} + r_{\text{coll}} + r_{\text{ps}} + \big(\phi(s_{t+1}) - \phi(s_t)\big) - 0.05\,\omega_t^2 - 0.025 + r_{\text{ac}}, \tag{1}$$

with $r_{\text{goal}} = +20$ on arrival, $r_{\text{coll}} = -20$ on collision, r_{ps} a proportional penalty for entering a 0.25 m personal-space zone around any pedestrian, ϕ a potential that rewards progress toward the goal, a spinning penalty and a per-step cost. The acoustic term r_{ac} is defined in Sec. V.

Algorithm. Proximal policy optimisation [7] with a shared recurrent actor-critic: 28 parallel environments, 30 steps per rollout, learning rate 5×10^{-5} with linear decay, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip 0.2, entropy coefficient 0.01.

IV. System

A. Architecture

Fig. 1 shows the pipeline. Each detected pedestrian contributes $[p_x, p_y, v_x, v_y]$ extended by five predicted future positions (14 values). The lidar scan is encoded by HEIGHT's 1D CNN. For occlusion inference a robot-centred 10×10 m occupancy grid of the pedestrians visible to a second 360° ray scan, with cells hidden behind the first object on each ray marked unknown, enters the PaS student encoder, whose 256-dimensional latent is appended to the robot state to form a 261-dimensional robot node. HEIGHT's attention layers relate the robot node to the pedestrian nodes, and its recurrent robot-node network combines the result with the obstacle features to select the action.

B. Intention prediction

Human nodes are extended with five predicted future positions at 0.1 s intervals. The predictions come from rolling each pedestrian's own ORCA policy and goal forward in the simulator without the robot (`sim.predict_method='truth'`), holding each pedestrian's current goal fixed. **This is privileged simulator state, not something a deployed robot could compute.** We adopted it as a near-best case for what a learned predictor could contribute; HEIGHT's released configuration uses no prediction. A learned predictor is scaffolded in the code but was not used for the results here.

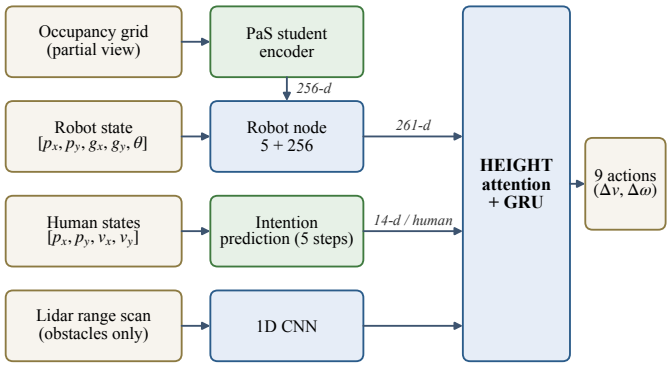


Fig. 1. System pipeline. Green blocks are our additions; blue blocks are HEIGHT's and unchanged, except that the robot node also carries the 256-dimensional occlusion latent. Obstacles reach the policy only through the lidar range scan.

C. Occlusion inference

We pre-train the teacher on ground-truth occupancy grids collected from 1,000 training episodes driven by uniformly random actions, then freeze its weights. The student is trained from scratch jointly with the policy under the distillation loss (coefficient 1.0). We audited the inference path and confirmed that the teacher is never called at test time; the student sees only the partial grid. The teacher is therefore not part of the deployed model.

D. Training

The combined policy (Stage 2) trained in two segments: 76 M steps from scratch, then a relaunch from that checkpoint which reset the optimiser, the learning-rate schedule and the step counter and ran a full 300 M-step schedule. Every arm evaluated here adds a further 50 M-step fine-tune to that shared checkpoint (Sec. V), for approximately 426 M environment steps in total. On a 40 GB MIG partition of an NVIDIA H100, at ∼829 environment steps per second, a 50 M-step fine-tune took roughly 17 hours.

E. Engineering notes

Two issues consumed substantial time and are worth recording. First, an early version of the observation code exposed the positions of occluded pedestrians to the policy and inflated its success until an audit of the observation pipeline found it (Sec. V). Second, Gym wrappers block access to underscore-prefixed attributes, and our use of `getattr` with a default value silently masked a metric-reporting failure. It surfaced only because we log a standard deviation for every metric: a counter with zero variance over 3,500 episodes is a bug, not a result.

V. Acoustic Extension

Beep. When the policy chooses a beep (at most once every 2 s), every pedestrian within 5 m receives a signal of strength $s(d) = \exp(-d^2/2\sigma^2)$, $\sigma = 3$ m. Each pedestrian is marked *compliant* at the start of an episode with probability 0.8; a

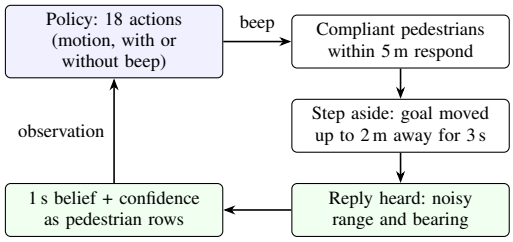


Fig. 2. The acoustic extension. The beep can clear a path (right) and, through pedestrians' replies, act as a sensor for people the robot cannot see (left, green). Ablations switch off the reply channel ("beep only") or its confidence feature.

compliant pedestrian answers a given beep with probability $0.8\,s(d)$. After a 0.2 s delay it moves its goal up to $2s(d)$ m directly away from the robot for 3 s, then returns to its original goal over 1 s (Fig. 2).

Listening. A responding pedestrian also replies. Within 8 m the robot obtains a range and bearing fix with noise $\sigma_r = 0.4$ m and $\sigma_\theta = 0.12$ rad, both multiplied by 2.5 when the speaker is occluded, since diffraction around an obstacle degrades localisation. Fixes over the 0.5 s reply are averaged, stored in world coordinates and kept for 1 s. A pedestrian the robot cannot see but has heard enters the policy as an extra pedestrian row. Every row carries a confidence feature, 1 for a visual detection and a value that decays with the fix's expected error and age for an audio estimate.

Reward. $r_{\text{ac}} = -0.1$ for each beep, plus 0.05 for each previously untriggered pedestrian the robot cannot see who is triggered by it, so the information bonus cannot be farmed by beeping at people already in view.

Training arms. All arms start from the shared Stage 2 checkpoint and are fine-tuned for 50 M steps with the same seed; only the acoustic switches differ. New input and output weights (the beep flag, the confidence column and the nine beep actions) are initialised fresh. The arms are: *no acoustics* (the matched baseline), *beep only* (no replies), *beep + listening* (full), and *beep + listening without the confidence feature*. All keep the privileged futures.

Development history. Earlier versions failed in instructive ways. In the first, the audio rows carried the exact positions of hidden pedestrians—a wall-hack that inflated success until an audit of the observation pipeline found it—and the information reward never fired, because the action was remapped to a motion before the reward checked for a beep; its results were discarded. With both fixed, a version whose audio fixes lasted only the 0.5 s of a reply reached 90% success on 500 test episodes. Holding each fix for 3 s without a confidence feature dropped success to 77%, with 23% collisions: with no motion model, a frozen estimate drifts by up to 1.5 m in 3 s, so the robot avoided ghosts and hit real obstacles. The final version holds fixes for 1 s, which caps the drift near the fix error, and adds the confidence feature (90% on the same 500 episodes). These $n = 500$ figures are development measurements only; the evaluation below uses 3,500 episodes per arm.

VI. Evaluation

A. Protocol

All rates are over $n = 3{,}500$ test episodes, which gives roughly 76% power to detect a 2-point difference in success between two fixed checkpoints at $\alpha = 0.05$; the conventional $n = 500$ has roughly 17% power for the same effect. The primary comparison is between arms that share the base checkpoint and training budget and differ only in the acoustic switches. HEIGHT's released checkpoint, evaluated in our codebase with our modules disabled, is shown for reference. Success contrasts use the pooled two-proportion z test with Wald 95% intervals. Because this report makes thirteen binary comparisons, we treat $p < 0.004$ (Bonferroni) as the conservative bar for any single one.

Base platform. Without acoustics, the platform scores 88.54% against 88.03% for HEIGHT's released checkpoint on the same test scenes ($+0.51$ points, $p = 0.50$): no detectable difference. Its full evaluation against HEIGHT is reported in the companion study [8] and not repeated here.

B. Acoustic ablation

Table I and Fig. 3 give the matched-budget comparison. Against the same fine-tune without acoustics, beeping raises success by $+0.94$ points ($[-0.52, +2.41]$, $p = 0.21$), and beeping with listening by $+1.06$ points ($[-0.40, +2.52]$, $p = 0.16$); neither is significant on this test set. Listening adds $+0.11$ points over beeping alone ($p = 0.88$), and removing the confidence feature costs 1.03 points ($p = 0.17$). The listening channel does work as a sensor—the robot localises 2.57 hidden pedestrians per episode, 0.34 per beep—but that information does not translate into measurably more successful navigation.

We evaluated the full and no-acoustics checkpoints on two further independent sets of 3,500 test scenes. The difference was $+1.06$, $+0.40$ and $+1.71$ points on the three sets, $+1.06$ points pooled over 10,500 episodes ($[+0.21, +1.90]$, $p = 0.014$). This replication is across test scenes, not across training runs: it shows that this particular pair of checkpoints differs, not that acoustic training reliably helps.

Beeping also shortens successful episodes by about 0.6 s (full: -0.61 s, Welch $t = -5.7$; beep only: -0.64 s, $t = -5.8$) and lowers the intrusion ratio, as one would expect when the pedestrians being beeped at walk away. The beep-only and full arms beep 5.1 and 7.6 times per episode (2.9 and 4.4 times per 100 steps).

C. Denser crowds and more obstacles

We tested the policies in the two harder conditions of HEIGHT's evaluation protocol: more pedestrians and more obstacles (Table II). Without acoustics, the platform stays level with HEIGHT ($+1.03$ points with more people, $p = 0.30$; -0.06 with more obstacles, $p = 0.95$). The beep helps most where the crowd is densest: with 10–14 pedestrians, beep-only is $+2.54$ points above the same policy without acoustics ($p = 0.0076$) and $+3.57$ points above HEIGHT ($[+1.69, +5.45]$, $p = 2 \times 10^{-4}$, the only contrast in this report that passes the Bonferroni bar). Listening again does not help:

TABLE I

ACOUSTIC ABLATION AT THE NOMINAL SETTING, $n = 3{,}500$ EPISODES PER ROW. ALL "OURS" ROWS ARE THE SAME 50 M-STEP FINE-TUNE FROM ONE SHARED CHECKPOINT WITH PRIVILEGED FUTURES, DIFFERING ONLY IN THE ACOUSTIC SWITCHES; THE FIRST TWO ROWS ARE REFERENCE ROWS FROM THE COMPANION STUDY [8]. COLLISION AND TIMEOUT COLUMNS ARE EPISODE COUNTS. NAVIGATION TIME IS OVER SUCCESSFUL EPISODES ONLY. "HEARD" COUNTS HIDDEN PEDESTRIANS LOCALISED BY THEIR REPLIES.

Arm	Success (%)	Pedestrian	Obstacle/wall	Timeout	Nav. time (s)	Beeps/episode	Heard/episode	Intrusion (%)
HEIGHT (released)	88.03	358	59	2	18.07 ± 4.36	–	–	5.66
Ours, no acoustics	88.54	336	56	9	18.21 ± 4.41	0	0	4.75
Ours, beep only	89.49	278	84	6	17.57 ± 4.27	5.06	0	4.22
Ours, beep + listening	89.60	293	69	2	17.60 ± 4.09	7.62	2.57	4.36
without confidence feature	88.57	318	76	6	17.40 ± 3.91	6.58	2.35	4.07

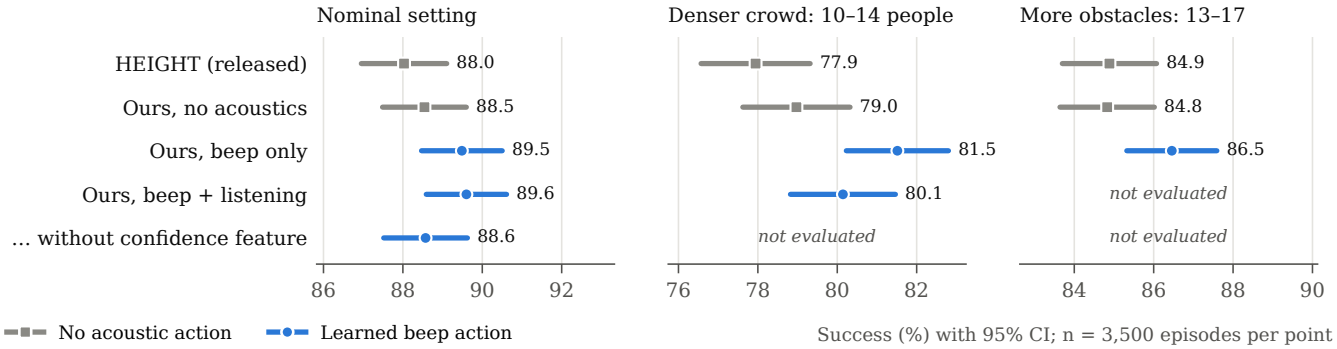


Fig. 3. Success with 95% intervals for every evaluated arm at the nominal setting and in the two harder settings. Each point is one checkpoint evaluated on 3,500 episodes; all policies were trained at the nominal setting.

TABLE II

OUT-OF-DISTRIBUTION SETTINGS, $n = 3{,}500$ EACH. ALL POLICIES WERE TRAINED WITH 5–9 PEDESTRIANS AND 8–12 OBSTACLES. COUNTS AS IN TABLE I.

Arm	Success (%)	Ped.	Obs./wall	Timeout
10–14 pedestrians				
HEIGHT	77.94	672	95	5
Ours, no acoustics	78.97	621	100	15
Ours, beep only	81.51	513	114	20
Ours, beep + listening	80.14	599	83	13
13–17 obstacles				
HEIGHT	84.89	280	236	13
Ours, no acoustics	84.83	306	191	34
Ours, beep only	86.46	215	226	33

TABLE III

MODEL SIZE. TRAINABLE ACTOR AND CRITIC PARAMETERS; THE FROZEN PAS TEACHER IS EXCLUDED BECAUSE IT IS NOT USED AT INFERENCE. THE ACOUSTIC VARIANTS ADD A FEW INPUT WEIGHTS AND NINE ACTION OUTPUTS TO "OURS" AND WERE NOT COUNTED SEPARATELY.

Model	Parameters	Checkpoint (MiB)
HEIGHT	478,618	1.84
Ours (prediction + occlusion)	1,236,058 (2.58×)	9.13

with more people the full arm is 1.37 points below beep-only ($p = 0.15$) and collides with pedestrians more often (+2.46 points, $p = 0.005$, short of the Bonferroni bar); we did not establish why. With more obstacles, beep-only is +1.63 points above no acoustics ($p = 0.052$).

D. Model size and compute

Prediction and occlusion inference cost 2.58× HEIGHT's parameters and 5× its checkpoint size (Table III). Both models are small by current standards—a few megabytes, well within the memory of an embedded GPU. Training is the dominant compute cost: each 50 M-step fine-tune took about 17 hours on a 40 GB partition of an H100, and the full combined recipe ran roughly 426 M environment steps. We did not measure on-board inference latency or power; that is the natural next measurement for a deployment-oriented version of this work.

VII. DISCUSSION

The beep works by moving people, not by informing the robot. Beeping alone captures the whole benefit of the acoustic extension at every density we tested, and the listening channel, although it localises hidden pedestrians as designed, adds nothing to success and may cost collisions in dense crowds. The effect of the beep is path-clearing, which is the mechanism of prior work [6], not the sensing mechanism we set out to test.

The benefit rests on a simulator assumption. Compliance is drawn independently of `visible_prob`, so about 64% of moving pedestrians (0.8×0.8) are modelled as unable to perceive the robot visually yet able to hear and obey its beep. That can be defended (someone looking at a phone may still react to a sound), but the size of the gain is governed by a compliance rate we chose rather than measured, and a

sensitivity sweep over that rate is the first experiment any follow-up should run. A beep every few seconds also has a social cost that the reward does not price beyond a small penalty.

Lessons for RL experiments. The first acoustic version looked strong because of an observation leak, not because of what it had learned. The fix for the stale-belief version changed two settings at once, so which one mattered could not be told until a separate arm removed the confidence feature. And at $n = 500$ a two-point difference has only 17% power, so none of our development numbers could have decided anything. Evaluate at adequate power, change one variable per run, audit the observation pipeline, and label privileged inputs in every table.

VIII. Limitations

Privileged information. Every "Ours" result, including all acoustic arms, gives the policy an ORCA forecast of each pedestrian computed from simulator state. No deployed robot has this, and whether the acoustic results hold with a learned predictor is untested.

One training seed per arm. Each arm was trained once; the three-scene-set replication varies test scenes, not training, so training-run variance is unmeasured.

Out-of-distribution tests. The denser settings were evaluated without retraining, on one checkpoint per arm, and the full arm was not run with more obstacles.

Simulated people. Pedestrians are ORCA agents and their response to a beep is a designed rule with a chosen compliance rate. How real people respond to a robot's sounds, and whether they would tolerate several beeps per crossing, is untested.

No hardware validation. All results are in simulation; inference latency and power on an embedded computer were not measured.

IX. Conclusion

We gave a graph-transformer navigation policy, trained with PPO, a learned acoustic action with a listening channel, and evaluated it in matched-budget ablations at a statistical power well above the field's norm. The beep raised success where crowds were dense, but through path-clearing under a compliance assumption, not through the information its echoes carried. The listening channel worked as a sensor yet did not improve navigation. Code is available online [9].

Code and data. The repository contains the simulator, policy, training and evaluation code, including the acoustic module (switched off by configuration in the released runs).

Use of AI tools. Anthropic's Claude was used to draft text and to write the code and analyses described here, and OpenAI's Codex to check the code, data and statistics. The author takes responsibility for the content.

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